

Jie-En (Matthew) Yao

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Education

University of Southern California

PhD in Computer Science; GPA: 3.95/4.00

Aug 2023 – May 2028

Los Angeles, CA

- Core Areas: Machine Learning, Deep Learning, Computer Vision, Image Processing, Generative Models, Medical AI

National Tsing Hua University

Bachelor of Science in Computer Science; GPA: 4.27/4.30; Class Rank: 1/142

Sep 2018 – Jun 2022

Hsinchu, Taiwan

Research Experience

University of Southern California and China Medical University Hospital

Research Assistant

Jan 2025 – Jan 2026

Los Angeles, CA

- **Topic: Coronary Angiography Interpretation and Stenosis Estimation**

- * Developed a clinically interpretable automated pipeline for medical image analysis
- * Implemented stenosis estimation algorithms to provide accurate vessel narrowing assessments, supporting cardiologists in data-driven clinical decision-making

MediaTek, Intelligence Vision Processing Team

AI Research Intern

Jan – Dec 2022

Hsinchu, Taiwan

- **Topic: Super-Resolution**

- * Prototyped, optimized, and iterated on advanced deep learning architectures, efficiently translating research ideas into high-quality experimental results
- * Developed a novel arbitrary-scale super-resolution model, achieving state-of-the-art performance and publication at CVPR

Publications

Jie-En Yao, et al., "HCL-FF: Hierarchical and Contrastive Learning for Forward-Forward Algorithm", accepted by IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2026

Jintang Xue, Jie-En Yao, et al., "Descrip3D: Enhancing Large Language Model-based 3D Scene Understanding with Object-Level Text Descriptions", accepted by IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026

Jie-En Yao, et al., "SHFE: Transparent And Green Image Classification Via Supervised Hierarchical Feature Extraction", accepted by IEEE International Conference on Image Processing Workshops (ICIPW), 2025

Jie-En Yao, et al., "Local Implicit Normalizing Flow for Arbitrary Scale Image Super-Resolution", accepted by IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2023

Ting-Hsuan Liao, Huang-Ru Liao, Jie-En Yao, et al., "ELDA: Using Edges to Have an Edge on Semantic Segmentation Based UDA", accepted by British Machine Vision Conf. (BMVC), 2022

Work Experience

Industrial Technology Research Institute

Data Science Intern at Computational Intelligence Technology Center

Jul – Sep 2020

Hsinchu, Taiwan

- Performed data preprocessing, feature engineering, and trained predictive Machine Learning models to forecast production quality, enabling data-driven quality control

Teaching Experience

Teaching Assistant of Principles of Programming for Data Science (Graduate level)

Aug - Dec 2025

- Led weekly lab sessions and office hours, supporting 50+ students develop Python and data science foundations

Teaching Assistant of Introduction to Programming (Undergrad level)

Jan - Jun 2021

- Supported curriculum organization and provided TA session to help students strengthen their C++ programming proficiency

Skills

Machine Learning / Deep Learning : PyTorch, NumPy, Scikit-Learn, XGBoost, Pandas

LLM / Generative Models : Hugging Face (Transformers, PEFT), LoRA, vLLM, Diffusion Models

Programming Language : Python, C++, C

Software Frameworks & Tools : Docker, Kubernetes, Git, Linux, CUDA